SKY130 · RV32I · 121 MHz · RELEASE v2.0

TRACE-V

A PicoRV32 RISC-V core on SKY130, built by agentic AI and verified with independent open-source tools

Traceable RISC-V: AI-built, Checked, Evidenced

8.25 ns clock 3 corners: ss / tt / ff 2 flows · 1 constraint file

License

NON-COMMERCIAL ACADEMIC INSTRUCTION AND RESEARCH USE

The released PicoRV32 sky130 design files and results are licensed for non-commercial academic instruction and research. Prior written permission is required for any other use.

Third-party components keep their own licenses

picorv32.v is licensed under ISC, © 2015 Claire Xenia Wolf. The SkyWater SKY130 PDK and standard-cell data are licensed under Apache-2.0 by SkyWater Technology Foundry and Google LLC.

Origin

The outputs in commercial/ were produced with EDA software licensed to Purdue University for academic instruction and research. The outputs in open/ were produced with open-source tools and verified in part with that software. The license keeps all outputs within that use.

Research collaboration or licensing inquiries about the framework are directed to Raisul Islam — raisul@purdue.edu.

Background: PicoRV32

THE CORE

- PicoRV32 is a size-optimized RISC-V CPU core by Claire Xenia Wolf, released under the ISC license [1].
- The core can be configured as RV32E, RV32I, RV32IC, RV32IM or RV32IMC [1].
- On Xilinx 7-series devices, the core runs at 250–450 MHz in 750–2,000 LUTs, with a CPI of about 4 [1].
- PicoRV32 is a general-purpose CPU, and RV32I is the 40-instruction RISC-V base integer ISA [2].

THIS BUILD: DEFAULT PARAMETERS

- The default parameters are used: RV32I, with 32 registers and counters [1].
- Multiply/divide, compressed instructions and interrupts are not included [1].
- Misaligned accesses and illegal instructions cause a trap [1].

Process: the SkyWater SKY130 open 130 nm PDK is used, with sky130_fd_sc_hd cells [3].

[1] github.com/YosysHQ/picorv32 · [2] RISC-V ISA Manual Vol. I (RISC-V International) · [3] github.com/google/skywater-pdk

The design

PicoRV32

The core is a 32-bit RISC-V (RV32I) design by Claire Xenia Wolf, under the ISC license.

SkyWater sky130A

The commercial layout holds 87,237 transistors in 8,273 standard cells of the open sky130_fd_sc_hd library.

8.25 ns · 121 MHz

The clock uncertainty is 0.25 ns at every corner.

`ss · 100 °C · 1.60 V`

`tt · 25 °C · 1.80 V`

`ff · -40 °C · 1.95 V`

COMMERCIAL IMPLEMENTATION (v1.0.1); THE OPEN FLOW IS ON SLIDE 05

Genus
synthesis
→
Innovus
place and route
→
Tempus
timing and SDF
→
Conformal
logic equivalence
→
Voltus
IR

Open checkers are used for physical signoff, so that anyone can re-verify the results: Magic DRC, KLayout DRC (sky130A_mr deck), Magic antenna, LVS with Magic and Netgen, and gate-level simulation with Icarus Verilog on the post-layout SDF.

The open flow: same RTL, same constraints

Release v2.0 adds a second build of the same RTL under the byte-identical constraint file: 8.25 ns with 0.25 ns uncertainty, maximum fanout 10, and a half-period I/O delay of 4.125 ns.

OPEN IMPLEMENTATION AND VERIFICATION · LIBRELANE 3.0.14, CLASSIC FLOW

Yosys
synthesis
→
OpenROAD
place and route
→
OpenSTA
timing
→
Conformal
logic equivalence
→
OpenROAD PSM
static IR

480.21 × 490.93 µm
die, 45% configured core utilization

122,928 µm²
standard-cell area

10,801
logic cells

144,143
transistors in logic cells

TOOL MODIFICATIONS

Implementation used OpenROAD with private modifications that are not published. The modifications concern timing repair and IR analysis. Every check an outsider runs uses unmodified open-source tools.

The open build was implemented with open_pdks 8afc8346, whose cell LEF and ss/tt libraries are byte-identical to those of 97d08448, which was used for all verification.

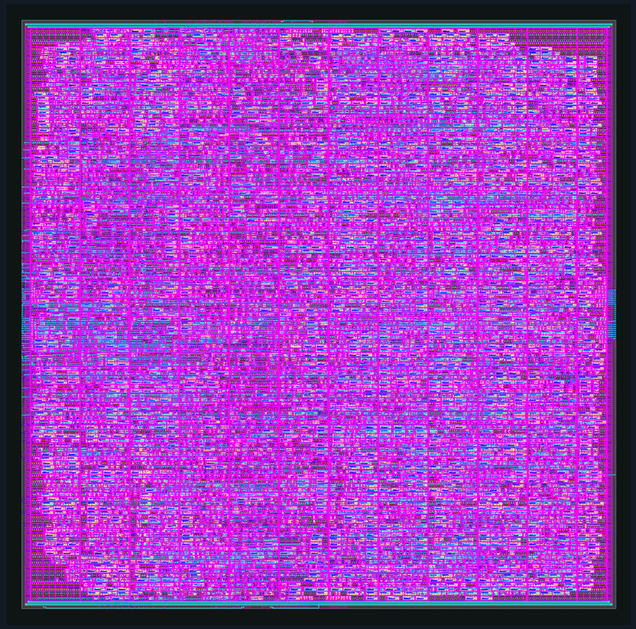
Equivalent tools in the two flows

Step	Commercial build (v1.0.1)	Open build (v2.0)
Logic synthesis	**Cadence Genus 25.11**	**Yosys 0.62 with ABC, in LibreLane 3.0.14**
Place, clock tree, route, timing repair	**Cadence Innovus 25.11**	**OpenROAD, modified, driven by LibreLane 3.0.14**
Extraction for timing	**Innovus, Cadence sky130 kit typical extraction file (nominal RC)**	**OpenRCX, in OpenROAD**
Signoff static timing	**Cadence Tempus 23.10**	**OpenSTA in OpenROAD, modified**
Static IR drop	**Cadence Voltus**	**OpenROAD PSM, modified**
Antenna repair	**Innovus NanoRoute, antenna margin**	**OpenROAD repair, then 2 diodes with LibreLane's stock ECO step**
Logic equivalence	**Cadence Conformal 24.10**	**Cadence Conformal 24.10 (same tool)**
Simulation with timing checks	**Cadence Xcelium 24.03**	**Cadence Xcelium 24.03 (same tool)**
Release verification, verify/	**Magic, KLayout, Netgen, OpenSTA, Icarus Verilog**	**the same unmodified tools**

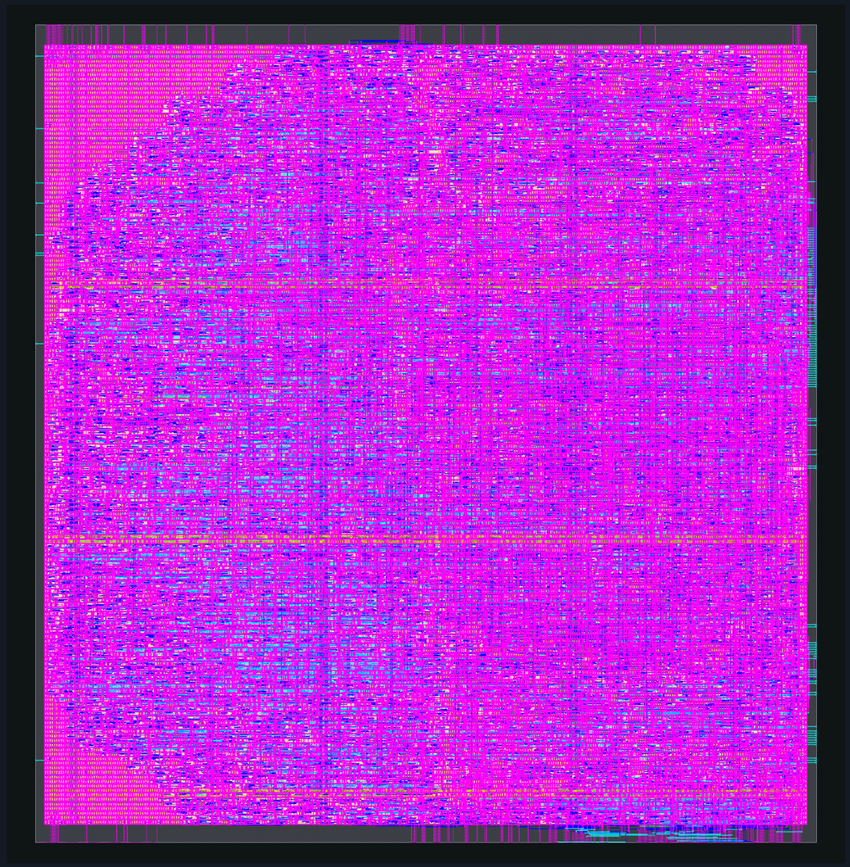
The upstream bases are OpenROAD dcf36133, OpenSTA 857316ff and LibreLane 3.0.14. The modifications are private and not published.

Routed layouts, same scale

COMMERCIAL (v1.0.1)
357.42 × 355.64 µm · 75,681 µm² cells (routed)
Genus + Innovus



OPEN (v2.0)
480.21 × 490.93 µm · 119,009 µm² cells + 3,919 µm² taps
Yosys + modified OpenROAD, 45% configured utilization



07 · KLayout renders of both final GDS files, drawn at the same scale

Physical and logical checks, both builds

Check	Commercial (v1.0.1)	Open (v2.0)
DRC, Magic	**0 violations**	**0 violations**
DRC, KLayout (sky130A_mr)	**0 findings**	**0 findings**
Process antenna, Magic	**0 violations (20,800 gates)**	**0 violations (30,900 gates)**
LVS, Magic + Netgen, from the GDS	**match; 8,273 devices, 8,492 nets**	**match; 11,237 devices, 10,944 nets**
LEC, RTL → synthesized netlist (Conformal)	**equivalent; 1,916 points**	**equivalent; 1,959 points (FSM encoding files and setup analysis)**
LEC, synthesized → routed netlist (Conformal)	**equivalent; 1,864 points**	**equivalent; 1,904 points**

DRC, antenna and LVS use the same unmodified open-source tools for both builds. LEC uses Conformal for both builds, and every run reports 0 non-equivalent, aborted or unknown points.

Timing, simulation and IR, both builds

Check	Commercial (v1.0.1)	Open (v2.0)
Setup, ss / tt / ff (shipped OpenSTA script)	**+0.280 / +2.403 / +2.964 ns**	**+0.129 / +2.028 / +2.706 ns**
Hold, ss / tt / ff (shipped OpenSTA script)	**+0.712 / +0.212 / +0.032 ns**	**+0.744 / +0.235 / +0.054 ns**
Flow's own signoff timer	**Tempus: setup ss +0.180 ns, hold ff +0.034 ns**	**OpenSTA in modified OpenROAD: setup max_ss +0.026 ns (max RC), hold min_ff +0.051 ns**
Slew / capacitance / fanout	**0 / 0 / 0**	**0 / 0 / 0**
Gate-level simulation, Icarus + SDF, 8.25 ns	**pass (RESULT 55); 4.125 ns control fails**	**pass (RESULT 55); 4.125 ns control fails**
Simulation with timing checks, Xcelium, 8.25 ns	**0 violations, 3,114 checks; 4 ns control: 76 violations**	**0 violations, 3,194 checks; 4 ns control: 2 violations**
Static IR, worst VPWR / VGND	**Voltus: 11.0 / 11.58 mV at 13.0 mW**	**PSM, modified: 5.55 / 5.57 mV at 22.0 mW**

Both builds use the same RTL and the same constraint file (8.25 ns, 0.25 ns uncertainty), and the slack includes the uncertainty. The IR budget is 80 mV. The IR figures come from different tools; the modified PSM on the commercial layout gives 12.09 / 12.07 mV.

The two builds side by side

	Commercial (v1.0.1)	Open (v2.0)
Clock / constraints	**8.25 ns, identical SDC**	**8.25 ns, identical SDC**
ss setup (shipped OpenSTA, nominal RC)	**+0.280 ns (OpenSTA) / +0.180 ns (signoff timer)**	**+0.129 ns**
ff hold	**+0.034 ns**	**+0.054 ns**
Die	**357.42 × 355.64 µm**	**480.21 × 490.93 µm (45% configured utilization)**
Logic transistors	**87,237**	**144,143**
DRC / antenna / LVS	**0 / 0 / match**	**0 / 0 / match**
Static IR (worst)	**11.0 / 11.58 mV (Voltus)**	**5.55 / 5.57 mV (PSM)**
Equivalence	**RTL → netlist → routed (Conformal)**	**RTL → netlist → routed (Conformal)**

The IR figures come from different tools. The PSM method, run on the commercial layout, gives 12.09 / 12.07 mV.

Step by step: the same block through both flows

Step	Commercial: tool, result	Open: tool, result
Synthesis	**Genus 25.11: 8,273 cells (final layout)**	**Yosys 0.62 + ABC: 10,801 logic cells (final layout)**
Place and route at 8.25 ns	**Innovus 25.11: closes**	**OpenROAD, modified: closes** stock OpenROAD, reference configuration, default SDC: −2.053 ns at max_ss, 3,104 slew violations
Timing, shipped OpenSTA script	**ss setup +0.280 ns · ff hold +0.032 ns**	**ss setup +0.129 ns · ff hold +0.054 ns**
Static IR, worst VPWR	**Cadence Voltus: 11.0 mV**	**PSM, modified: 5.55 mV** on the commercial layout 12.09 mV; unmodified 46.4 mV
Logic equivalence	**Conformal 24.10: RTL → routed proven**	**Conformal 24.10: RTL → routed proven**
Simulation, timing checks	**Xcelium 24.03: 0 violations, 3,114 checks**	**Xcelium 24.03: 0 violations, 3,194 checks**
DRC / antenna / LVS	**Magic, KLayout, Netgen: 0 / 0 / match**	**Magic, KLayout, Netgen: 0 / 0 / match**

Both builds use the same RTL and the same constraint file. Conformal and Xcelium are used for both builds: the open equivalence method did not prove RTL to netlist for this design (586 of 1,864 key points; the x-aware sequential proof was undecided after 2 h), and Icarus Verilog does not evaluate timing checks. The open implementation uses no commercial tool.

MODIFIED OPEN-SOURCE TOOLS

Where the open-source tools were changed

OpenROAD Resizer: repair

After routing, electrical and timing repair use the extracted parasitics, including zero-length shunt capacitances. The replaced estimate under-counted wire load 2.2 to 2.4 times on two measured nets. Driver sizing uses the corner model and the SDC slew limit.

OpenSTA in OpenROAD: parasitics

The SPEF reader and parasitic store give the Resizer the extracted RC network in a form its buffering model can use; constraint checks ignore empty exception entries.

OpenROAD PSM: static IR

Voltage sources can sit on each power pin's own routing layer as well as on the top layer.

LibreLane: flow integration

A post-route loop repairs electrical, setup and hold violations against fresh extraction, and the flow fails when a stated setup target is missed.

Yosys and ABC, Magic, KLayout, Netgen and Icarus Verilog were not modified. The changes are private and not published.

Stock against modified, on the same block

8.25 ns, same RTL, same configuration	Stock LibreLane 3.0.14 / OpenROAD	Modified OpenROAD
Setup, max_ss / nom_ss	**−2.053 / −1.672 ns**	**+0.018 / +0.147 ns**
Slew / cap / fanout violations, max_ss	**3,104 / 33 / 30**	**0 / 0 / 0**
Hold	**fails at max_ss (−0.104 ns)**	**≥ 0 at all corners (+0.075 ns min_ff)**
Flow exit	**errors**	**clean**

Static IR on the commercial layout	VPWR / VGND worst
Cadence Voltus	**11.0 / 11.58 mV**
PSM, unmodified, default sources	**46.4 / 47.0 mV**
PSM, unmodified, sources at Voltus's pin locations	**36.8 / 34.8 mV**
PSM, modified, sources on each pin's layer	**12.09 / 12.07 mV**

The timing rows use the LibreLane reference configuration with the default SDC. The release build closes under the commercial SDC with every check clean (max_ss +0.026 ns).

Software developed for this work

Modified open-source tools

The changes are in OpenROAD Resizer repair, the OpenSTA parasitics inside OpenROAD, OpenROAD PSM and the LibreLane flow integration. They are private and not published.

Adapters used for this release

The commercial flow uses Genus, Innovus, Tempus, Voltus and Conformal adapters behind an evidence registrar and ledger. The open flow uses recorded LibreLane runs, the Conformal set-up for the open netlist, PSM runs with controls and the ECO diode run. Both use the shipped verify/ scripts.

Developed alongside

An open equivalence adapter over Yosys and ABC was run on this design and gives only the negative result on slide 11. An open static-IR adapter over PSM, and a LibreLane runner and pre-flight plugin that read every setting back, are not the source of any released number.

Every cited result comes from a run whose inputs and outputs are frozen with checksums, and each verify script was checked to fail on a deliberately damaged input.

How the platform works

01

Requirements and implementation

The requirements were set once by the designer; synthesis, place and route, signoff and debugging were carried out by agentic AI.

02

Traceable figures

Every figure is traceable to the run and input files that produced it.

03

75 frozen evidence files (commercial build)

Cited evidence is frozen and checksummed.

04

Every physical, DRV and simulation check has a control

Every physical, DRV and simulation check was shown to fail on a known-bad input before it was trusted.

05

No silent passes

A check that cannot run is recorded as a failure.

06

Independent checkers

Layouts built by Innovus are checked by Magic, KLayout and Netgen; LEC links the routed netlist back to the RTL.

DEFECTS FOUND DURING SIGNOFF

Commercial flow: six defects that passed the tools' own reports

Each result was read back from the design and compared with the requirement.

Results that should agree were compared.

Each check was shown to fail on a known-bad input before it was trusted.

Defect 1: the fanout limit was ignored

A maximum fanout of 10 was set in the SDC. Innovus finished with a 240-pin fanout violation, and the flow completed normally.

fanout violation

240 → 0

clock-tree buffers

23 → 174

insertion delay (CTS average)

0.46 → 0.95 ns

Gray bar: before the fix. Amber bar: after.

HOW IT WAS FOUND

The fanout count was read back from the routed design and compared with the limit declared in the SDC. Signoff was refused at 240 pins over, although Innovus reported success.

The two settings that Innovus needs (optimizer and clock-tree fanout repair) were found by agentic AI, the route was re-run, and the count is now 0.

Defect 2: STA passed, simulation trapped

`mem_ready → reg_op1`

-1.87 ns

The interface was timed as the testbench drives it: memory is driven on the falling edge, which leaves half a cycle.

1. After the fanout fix, setup was met by STA at 8 ns, but the post-layout simulation trapped at 8 ns.
2. An instruction that had been given correctly was misread by the core.
3. The cause was the absence of input and output delays in the SDC, so the memory interface was given a full cycle by STA instead of the testbench's half cycle.

HOW IT WAS FOUND

Two results contradicted each other: STA met, simulation failed. The STA pass was not accepted. The wrong transaction was traced by agentic AI, the interface was timed the way the testbench drives it, and the constraint was fixed instead of the test.

After the re-run, the simulation passes at 8 ns and the input-port hold violations (down to −0.92 ns before) are gone.

Defects 3 and 4: faults in the checks themselves

DEFECT 3

The gate-level simulation was all-X

The supplies of the power-aware cell models were unconnected, so every simulation failed, at every clock.

CONTROL

The defect was found by a zero-delay positive control: the same netlist with no delays must pass.

DEFECT 4

The antenna check could not fail

The initial negative control contained a diode that was not needed, so the check passed it.

FIX

The control was replaced by a frozen known-bad layout that must fail.

HOW THEY WERE FOUND

Every check carries a control that it must pass or fail. When a control misbehaved, the check was marked unproven instead of clean, and the check itself was repaired by agentic AI before any result from it was trusted.

Defects 5 and 6: found in the 8.25 ns re-run

DEFECT 5

A cell excluded from synthesis was re-inserted by place and route.

The and2_0 cell was excluded in Genus, but Innovus restructuring re-inserted it. One instance abutted a neighbor and produced a KLayout npc.2 spacing violation (0.16 µm against 0.27 µm), while Magic DRC was clean.

FOUND AND FIXED

The defect was found by the second DRC deck. It was fixed by applying the exclusion in Innovus as well, with a post-route count of excluded cells that must be zero.

DEFECT 6

An antenna violation remained after the router's repair.

One net was left at a ratio of 403.85 against a limit of 400.

FOUND AND FIXED

The defect was found by the open antenna check. It was fixed by repairing to 96% of the limit.

Defects found and fixed in the v2.0 work

CONSTRAINT MISMATCH

The open flow initially used LibreLane's default SDC.

Its I/O delay was 1.65 ns. Timed under the commercial SDC, that build failed ss setup by −2.294 ns at the input resetn.

FOUND AND FIXED

The build was re-run under the identical SDC, and it closes.

FRAGILE CLOSURE

A closure did not survive a small placement change.

Before the timing-repair modification, an 8.25 ns closure at +0.019 ns fell to −0.195 ns when the wire-length weight moved from 0.25 to 0.3.

FOUND AND FIXED

After the modification, the same perturbation passes at +0.039 ns.

ANTENNA CHECKERS

OpenROAD reported 0 antenna violations; Magic flagged 2 to 3 gates per layout.

The cause and the proof are on the next slide.

FOUND AND FIXED

On the release layout, two diodes were added with a stock ECO step: 492 → 494 diodes, +5 µm².

EQUIVALENCE SETUP

A plain key-point comparison could not prove equivalence.

Re-encoded state machines and constant registers left 1,268, then 1,200, non-equivalent points.

FOUND AND FIXED

Three setup steps resolve it, with a control (slide 23).

Verification environment: with a newer PDK than the pinned open_pdks 97d08448, the diode cells failed LVS, so the README pins that version.

Two antenna checkers disagreed

OpenROAD reported 0 antenna violations on the routed open builds. Magic, which the shipped scripts run, flagged 3 gates on an earlier candidate layout and 2 on the release layout.

1,662.6 vs 1,666.5
net _06165_: computed ratio against the ratio Magic reports

0
Magic violations after the metal4 bridge is erased in a copy

492 → 494
diodes after a stock ECO step; +5 µm²

+0.027 → +0.026 ns
max_ss setup slack after the fix

CAUSE

Magic counts metal3 that reaches a gate through metal4 alone as connected at the metal3 step. On net _06165_, the computed ratio of 1,662.6 agrees closely with the 1,666.5 that Magic reports, and erasing the metal4 bridge in a copy of the layout takes Magic to 0.

FIX

Two diodes were inserted with LibreLane's stock Odb.InsertECODiodes step, and the Magic check reports 0 violations. A patched mode that added 2,098 diodes (+9.7% area) was considered and not used.

Equivalence of the open build: three setup steps

Yosys re-encodes state machines and leaves registers that the default configuration never uses. A plain key-point comparison against the RTL cannot prove equivalence. Conformal LEC 24.10 proves it after three steps.

STEP 1

Flip-flops renamed after their RTL registers.

The renaming script asserts that cells and connections are identical.

RESULT

Names change; cells and connections stay identical.

STEP 2

FSM encoding files for two state registers.

cpu_state: 8-bit one-hot to 7-bit one-hot.
mem_wordsize: 2-bit binary to 3-bit one-hot.

CONTROL

Without these files, 1,246 points are non-equivalent.

STEP 3

Conformal setup analysis.

77 RTL registers were merged (duplicates and registers the default configuration never uses), and 5 were treated as constants.

RESULT

Equivalent: 1,959 points, 0 non-equivalent, aborted or unknown.

The Yosys netlist is also equivalent to the routed netlist of the final layout: 1,904 points, 0 non-equivalent.

What it cost

1 Oct → 7 Oct 2026

The calendar time runs from the start of the commercial flow on the open sky130 PDK to block signoff (7 Oct, 16:15 EDT).

38 min · 45 min

One full signoff pass takes 38 min of compute and 45 min end to end.

62 jobs · 5.1 h

62 cluster jobs ran from 1 Oct through the 8 ns signoff, with 5.1 h of total wall time. The 8.25 ns re-run added 21 cluster jobs and 2.0 h of wall time.

2,907 calls

Agentic AI usage was 2,907 calls, 1.26 M output tokens, 0.89 B cached input tokens read, 32 M written to cache and about 6.5 k uncached.

> About 3.1 hours of designer interaction were spent over 7 days, in 86 messages. The requirements were set and each decision was approved by the designer. Implementation, verification, debugging and fixes were carried out by agentic AI.

Times are measured from message timestamps, with gaps under 20 minutes counted as work. Token and time figures are upper bounds, because parallel work outside this design was included.

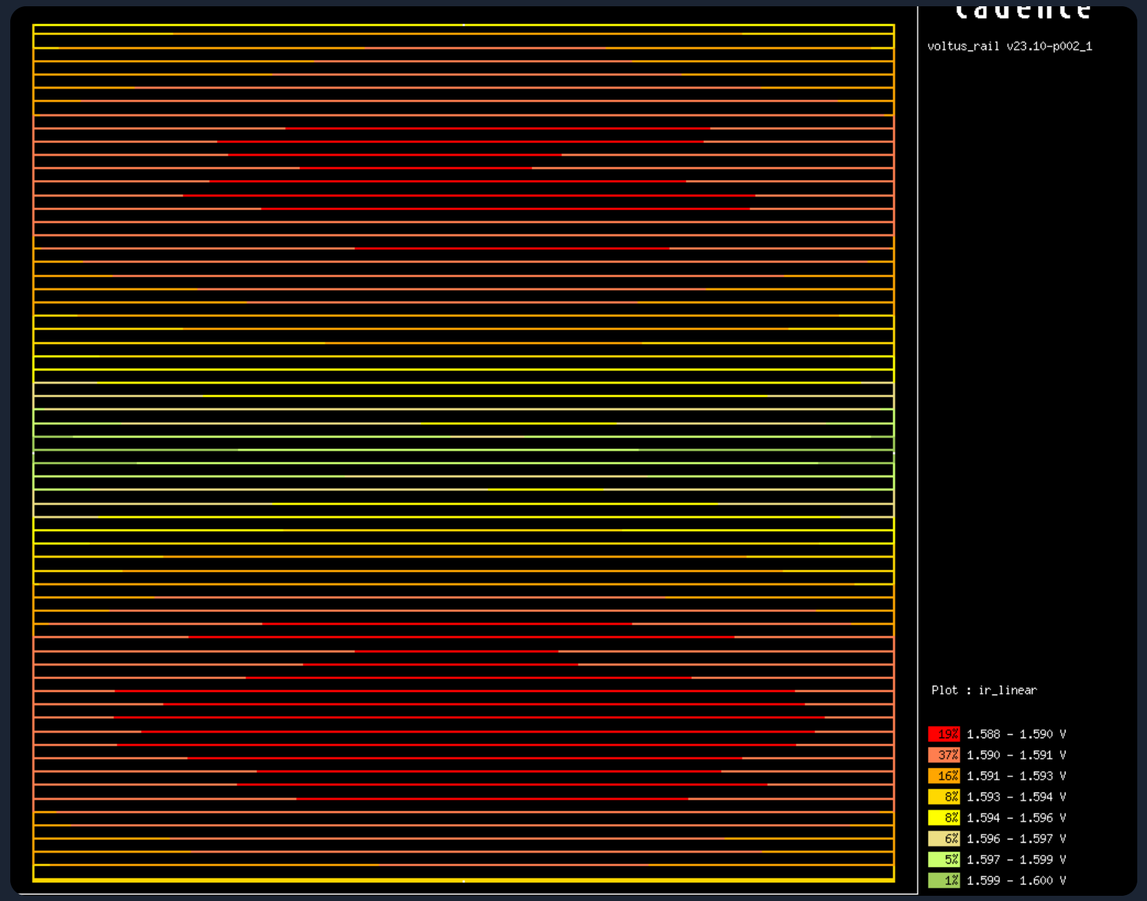
Commercial build: static IR drop, 11 mV on VPWR

11 / 11.6 mV
maximum drop, VPWR / VGND

80 mV each
budget, 5% of 1.6 V

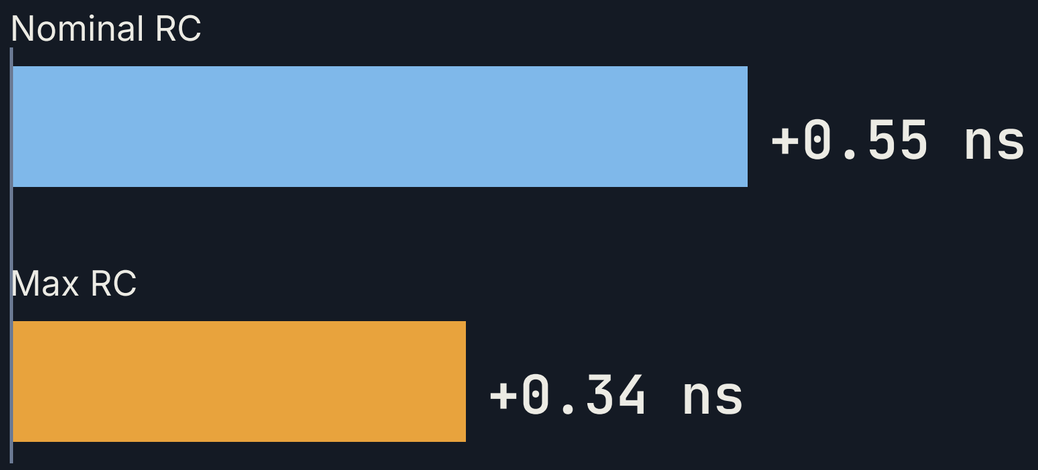
1.588–1.600 V
legend range of the VPWR map, 8.25 ns route

The analysis is static and vectorless, with 20% input activity at 8.25 ns.



Commercial build: RC cross-check

The cross-check was run with an independent extractor, OpenRCX, and with OpenSTA: ss setup at 8.25 ns.



Max RC costs 0.21 ns.

The ff hold with min RC is +0.03 ns, which is met.

Signoff extraction is at nominal RC, because the Cadence sky130 kit provides a typical extraction file.

This is an estimate combining two extractors: the 0.21 ns applied to the +0.180 ns signoff margin gives about −0.03 ns, so **about 8.3 ns** would be needed at max RC.

Limitations

1. Commercial build: signoff is at nominal RC; the max-RC estimate is given on slide 26.
2. Both builds: the I/O timing is set to the testbench's half-cycle memory protocol.
3. Commercial build: the margins are thin, ss setup +0.180 ns and ff hold +0.034 ns, after 0.25 ns uncertainty.
4. The post-layout simulation runs one 9-instruction program covering 6 of 40 RV32I opcodes; LEC and STA cover the whole design.
5. Open build: static IR comes from OpenROAD PSM with private modifications, and no Voltus run exists on the open layout. Its Conformal and Xcelium results are not regenerated by the open scripts.

Scope: Packaging, electromigration, metal fill, pad frame, foundry precheck and dynamic IR were outside the scope of this block-level release.

Sources

[1] C. X. Wolf, PicoRV32, github.com/YosysHQ/picorv32.

[2] A. Waterman, K. Asanović (eds.), The RISC-V Instruction Set Manual, Vol. I, RISC-V International.

[3] SkyWater SKY130 PDK, github.com/google/skywater-pdk, built with open_pdks 1.0.471 (github.com/RTimothyEdwards/open_pdks).

Tools Cadence Genus, Innovus, Tempus, Conformal, Voltus, Xcelium; open flow Yosys, OpenROAD, LibreLane 3.0.14; open checkers Magic, KLayout, Netgen, OpenSTA, OpenRCX, Icarus Verilog.

What this release covers

Each build was signed off at three corners with 0.25 ns clock uncertainty.

Each verify script was checked to fail on a deliberately damaged input.

DRC was run with two independent decks, LVS against the routed netlist, and logic equivalence at both synthesis and routing.

Every reported figure is traceable to frozen, checksummed evidence.

Post-layout gate-level simulation was run with flop timing checks, and static IR was analyzed.

The physical, timing and simulation checks can be reproduced from either public package with open-source tools.